# Sharp-Feature Preservation in Neural Signed Distance Fields under a Fixed Compute Budget

Periklis Lazarides

P-number: P2669504

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De Montfort University, Leicester

Supervisor: Dr. Conor Fahy

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# Declaration

I declare that this dissertation is my own work, except where otherwise acknowledged, and that all sources used have been properly cited. The work has not been submitted, in whole or in part, for any other award at this or any other institution.

Periklis Lazarides
P2669504
September 2026

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I would like to thank my supervisor, Dr Conor Fahy, for guiding me, especially in the early stages of figuring out the main technique that I would use and explore for my dissertation.

The test mesh is the Male Base Mesh by nixor, credited to Paul Chen, used under a Personal Use License from Free3D.

# Abstract

Sharp features are often softened during polygon mesh reconstruction because the smoothness assumption that makes classical methods resistant to noise can also remove real geometric discontinuities. Neural signed distance fields instead rely on learned representations, and periodic sine activations have been proposed because they can represent high-frequency detail more readily than piecewise linear networks. The claim is widely cited and rarely tested under control. This project compares a SIREN against a ReLU multilayer perceptron of identical capacity, both fitted to the same cached sample set for the same number of iterations and meshed at the same resolution, so that the activation function is the only variable. Quality is measured by symmetric Chamfer distance, by directed accuracy and completeness at the median, 90th and 99th percentiles, by uncovered surface fraction across ten tolerances, and by normal consistency. At the matched learning rate SIREN is better on almost every measure, halving symmetric Chamfer distance and reducing accuracy at the 99th percentile by a factor of ten. Against a baseline at the better of the two learning rates tested, the picture is mixed. The two are separated by 0.08 per cent on symmetric Chamfer distance while their accuracy tails differ by a factor of 3.3, because the aggregate sums two directed errors that move in opposite directions and cancel. Failures are contiguous and concentrated on the same high curvature regions in every configuration, so the activation function changes their severity and not their location.

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# 1. Introduction

## 1.1 Context and motivation

Turning an unstructured point cloud into a usable polygon mesh is among the oldest problems in computer graphics and one of the least settled. The classical pipeline fits an implicit function over space and extracts its zero-level set with an isosurfacing algorithm, and screened Poisson reconstruction remains the method most production tools reach for. These approaches are fast and robust to noise. They also share a characteristic failure, which is that they smooth. A field fitted under a smoothness prior rounds any feature whose curvature exceeds what the prior expects, and a mesh extracted on a uniform grid cannot represent a discontinuity falling between grid points. Creases, corners and thin structures are therefore the first things lost.

This matters in practice. Virtualized geometry systems such as Nanite assume that source assets carry detail far beyond what any frame rasterises, which moves the quality ceiling upstream onto reconstruction. An asset arriving with its edges already rounded cannot be repaired by any amount of downstream level of detail machinery, and the manual alternative of retopologising by hand does not scale.

A neural signed distance field is an interesting candidate because its prior is learned rather than imposed. A network is trained to output the signed distance to a surface, can be queried anywhere in space, and is meshed with marching cubes exactly as an analytic field would be. In principle it does not need to round a feature it has reason not to round, and Sitzmann et al. (2020) argue that the activation function is what decides whether it does.

## 1.2 Problem statement

A coordinate network's capacity for high frequency geometry is largely set by its activation function. A ReLU network is piecewise linear and biased toward low frequency solutions, which is the bias that rounds an edge. Sitzmann et al. proposed periodic sine activations on the grounds that they represent high frequency detail and its derivatives more readily, and the reported results are visually sharper. The claim is widely repeated and rarely tested under control.

Two weaknesses recur in the supporting evidence. Comparisons typically vary architecture and training configuration together, so a quality difference cannot be attributed to the activation function specifically. The metric reported is usually a mean or symmetric Chamfer distance, which averages over the whole surface and is therefore insensitive to localised feature loss, which is the exact failure this project cares about.

## 1.3 Aim and objectives

The aim is to determine whether the choice of activation function materially affects sharp feature preservation in signed distance field reconstruction, by comparing a SIREN against a ReLU multilayer perceptron under a matched compute budget and evaluating with metrics sensitive to localised rather than average error.

1. Implement a reconstruction pipeline carrying a triangle mesh through distance sampling, network fitting, marching cubes extraction and evaluation, deterministic under a fixed seed.
2. Implement both architectures identical in width, depth, parameter count and input encoding, holding sample set, iteration count, batch size and extraction resolution constant so that the activation is the only variable.
3. Define an evaluation protocol reporting accuracy and completeness separately at the median, 90th and 99th percentiles and the maximum, with uncovered fraction and F score across a range of tolerances and a measure of normal consistency.
4. Run the controlled comparison and report the full metric set, including any disagreement between aggregate and tail metrics.
5. Characterise where each reconstruction fails, and whether the failing regions are shared between architectures.

## 1.4 Research questions and hypothesis

RQ1. Under a fixed compute budget of matched sample count, iteration count, learning rate and extraction resolution, does a SIREN reconstruct sharp geometric features more accurately than a ReLU baseline?

RQ2. Where does each architecture fail, are those failures diffuse or concentrated in contiguous regions, and what do they reveal about the trade off each architecture makes?

RQ3. How informative are standard reconstruction metrics for this question, and what does any divergence between aggregate and tail measures imply about how neural signed distance reconstruction should be evaluated?

The hypothesis tested is that under matched conditions the SIREN reconstructs high curvature features with lower error than the ReLU baseline, and that this advantage is visible in tail measures while being obscured in aggregate ones.

## 1.5 Contributions

- A complete reconstruction pipeline covering sampling, training of both architectures, extraction and evaluation.
- A controlled comparison of the two under matched training configuration.
- A multiple threshold evaluation protocol, and evidence that aggregate and tail metrics disagree on which architecture is better.
- An account of a mixed result, including three defects whose correction changed the reported ordering before the final results were obtained.

## 1.6 Relationship to the Terms of Reference

The Terms of Reference, reproduced in Appendix A, committed to a reconstruction pipeline, a controlled comparison of two architectures, and an evaluation of the results. The pipeline is described in Section 3.2 and shown in Figure 3.1, the two architectures in Section 3.4, and the comparison in Chapter 4. All deliverables recorded in the Terms of Reference were produced, with the deviations in scheduling described in Chapter 6.

## 1.7 Dissertation structure

Chapter 2 reviews classical and neural reconstruction, activation functions and evaluation practice. Chapter 3 sets out the experimental design. Chapter 4 reports results. Chapter 5 answers the research questions and evaluates the artefact. Chapter 6 reviews the plan against what happened, and Chapter 7 concludes.

# 2. Literature Review

## 2.1 Classical surface reconstruction

The standard route from an implicit function to a polygon mesh is marching cubes (Lorensen and Cline, 1987), which samples the function on a regular grid and, for each cell containing a sign change, emits triangles from a lookup table indexed by the signs at the eight corners, placing vertices along cell edges by linear interpolation. It is fast, parallel and agnostic to how the field was produced, which is why it remains the default nearly four decades later. Its weaknesses are structural. The original table resolves several configurations ambiguously, so a shared face can be triangulated inconsistently and leave a hole, addressed later by the asymptotic decider of Nielson and Hamann (1991). More consequentially here, vertices are confined to cell edges, so a feature falling between grid points is unrepresentable and is instead approximated in a stepped form. Grid resolution therefore places a ceiling on recoverable detail independent of the quality of the field, which is the confound Section 5.6 must acknowledge.

Producing that field from an unstructured point cloud is the harder half, and Poisson surface reconstruction (Kazhdan, Bolitho and Hoppe, 2006) is the dominant answer, solving for an indicator function whose gradient best matches a vector field derived from oriented input normals. Screened Poisson reconstruction (Kazhdan and Hoppe, 2013) adds a term constraining the solution to interpolate the sample positions, sharpening results at modest cost. The formulation is robust to noise precisely because it is a global least squares fit, and that robustness is the source of its characteristic failure, since a smooth solution is preferred by construction and a crease supported by few samples is regularised away. Both methods also require oriented normals, which a raw scan does not supply, and which must be estimated by a procedure that is itself least reliable in the high curvature regions where the reconstruction most needs them.

A body of work addresses feature loss by changing the extraction stage rather than the field. Extended marching cubes (Kobbelt et al., 2001) detects cells whose normals indicate a crease and inserts a vertex at the intersection of the local tangent planes, and dual contouring (Ju et al., 2002) generalises this by placing one vertex per cell at the minimiser of a quadratic error function built from positions and normals at the sign changes. Both reproduce creases that marching cubes cannot express. The requirement is the limitation, since both consume normal information at the point of extraction, which is available when the implicit function is analytic and unreliable when it is fitted to a noisy scan. They relocate the problem rather than solve it, the accuracy of a recovered feature depending on the accuracy of the normals used to place it.

The smoothing is therefore a property of the prior rather than a defect in it, since the same assumption that supplies reliability to noise and makes the problem well posed under incomplete data is what rounds the feature. Feature preserving extraction compensates downstream when normal information exists. The alternative, and the one this project examines, is to change the prior itself.

## 2.2 Neural implicit representations

DeepSDF (Park et al., 2019) replaced the analytic prior with a learned one, training a multilayer perceptron to map a coordinate, optionally concatenated with a shape latent code, to a signed distance. Two choices from that work carry into this project. Distances are truncated to a narrow band before the loss is computed, since capacity spent far from the zero-level set is wasted where no isosurfacing algorithm reads it, and an L1 loss is preferred to a squared error because the samples concentrate near the surface, where a few difficult points would otherwise dominate. The reported quality, however, is a Chamfer distance averaged across shape categories in a learned latent space, which says little about how a single network represents a single surface.

Occupancy Networks (Mescheder et al., 2019) arrived independently at a related formulation, predicting binary occupancy rather than distance. An occupancy field is a classifier output with no metric meaning away from the boundary, whereas a distance field is interpretable everywhere and can be queried by sphere tracing. The truncation used here sacrifices most of that advantage, a limitation rather than a neutral implementation detail.

A second line removes the need for ground truth distances, the practical obstacle to applying these methods to real scans. Implicit geometric regularisation (Gropp et al., 2020) adds an Eikonal term penalising deviation of the gradient norm from unity. Neural Pull (Ma et al., 2021) uses the predicted value and gradient to pull query points onto the surface, needing no distance labels at all. DiGS (Ben-Shabat, Koneputugodage and Gould, 2022) targets unoriented point clouds with a divergence term on the gradient field, arguing that second order information can substitute for the normals earlier methods require. These are more applicable than the supervised formulation used here, and they also introduce regularisation weights that interact with architecture, which is the confound a controlled comparison must avoid. Supervising on exact distances is a restriction on generality that buys the ability to attribute a difference in output to the activation function alone.

A third line accepts the coordinate network but changes what is fed into it. NGLOD (Takikawa et al., 2021) stores learned feature vectors at the corners of a sparse voxel octree and interpolates them as input to a small decoder, and Instant NGP (Müller et al., 2022) replaces the octree with a multiresolution hash table, reducing training from hours to seconds. Both recover high frequency geometry a plain coordinate network of comparable size cannot, and both do so through the input representation while leaving the activation conventional. This is

the strongest counterargument to the framing of the present project, since it suggests that where detail comes from is a question about encoding rather than nonlinearity, and it means the unencoded baseline used here is not representative of what a ReLU based method can achieve.

What is common across this work is the manner of demonstration rather than the results. Architectural claims are supported by renders chosen to display the difference, or by a mean Chamfer distance over a test set, while the compared systems differ in training length, sample count, capacity and extraction resolution at once. Few hold the budget fixed and fewer report where the error is concentrated, so attribution to any single design decision is rarely supported by the evidence presented.

## 2.3 Activation functions and spectral bias

The theoretical account of why a plain ReLU network struggles with detail is spectral bias. Rahaman et al. (2019) showed that such networks fit low frequency components of a target function first and high frequency components slowly or not at all, a consequence of the piecewise linear structure and of the eigenspectrum of the associated kernel. For a coordinate network this is a statement about which features survive. A crease is a high frequency component of the distance field, so the bias that stabilises training also rounds the edge, and the neural method inherits the classical failure of Section 2.1 by a different route.

Two responses exist. The first modifies the input, mapping coordinates through sinusoids of increasing frequency before the first layer. Tancik et al. (2020) analysed this through the neural tangent kernel and showed that a Fourier feature mapping controls the bandwidth the network can fit, the positional encoding of NeRF (Mildenhall et al., 2020) being the widely adopted instance. The second modifies the network, and SIREN (Sitzmann et al., 2020) replaces every activation with a sine scaled by a factor omega_0. Because the derivative of a sine is a sine, the argument extends to derivatives, which matters for a field whose gradient carries the surface normal. The factor omega_0, set to 30 in that work, multiplies each pre-activation and selects the frequency band a layer operates in, and the accompanying initialisation scales hidden weights by the reciprocal of omega_0 so that pre-activations retain a fixed distribution through depth. That scheme is presented as necessary rather than incidental, implying a network tightly coupled to a few scale parameters.

The prediction that follows is unambiguous. On geometry containing high frequency detail a periodic activation network should outperform a ReLU network of equal capacity, and the claim is repeated widely enough to function as established fact. It has been tested far less often than cited, and almost never with the training configuration held constant across the compared architectures. A mixed or null result under those conditions is therefore worth

reporting, since it bears on how much of the advantage is representational and how much is an artefact of the comparison.

## 2.4 Evaluation of reconstruction quality

Chamfer distance is the standard quantitative measure in this literature. In symmetric form it sums two directed terms, the mean distance from each reconstructed sample to its nearest ground truth neighbour and the mean in the opposite direction. It requires no correspondence and is reported by DeepSDF, Occupancy Networks and most work following them, which makes it the only figure available for comparison across papers. Its properties as a statistic receive less attention than its ubiquity would justify.

Two of those properties matter here. It is a mean, and a mean over a densely sampled surface is dominated by the regions that are correct, so a reconstruction accurate over ninety nine per cent of its area and catastrophically wrong on the rest scores close to a uniformly good one, though a practitioner would reject the first and accept the second. It is also a sum of two directed quantities measuring different failures, so a method that fabricates geometry and one that omits it can produce identical scores. The directed terms are sometimes reported separately as accuracy and completeness, a formulation borrowed from precision and recall that is standard in stereo benchmarking and optional in the neural implicit literature.

The benchmarking literature for reconstruction from multiple views has largely moved past means for this reason. The DTU (Jensen et al., 2014) and Tanks and Temples (Knapitsch et al., 2017) benchmarks report precision and recall at a distance tolerance and combine them into an F score, asking what fraction of the surface is correct to within a stated threshold rather than how wrong it is on average. This matches how reconstructions are used, since a downstream stage enforces a tolerance rather than a mean. The cost is a parameter, because the ranking depends on the chosen tolerance and a single value conceals that dependence, which argues for reporting a curve rather than a point.

Percentile reporting is the natural complement and is uncommon in this field. A ninety ninth percentile describes the worst region of a surface and the ratio of mean to median describes the skew of the error distribution, both more informative than the mean for work concerned with features rather than bulk fidelity. Normal consistency is a further complement, since a surface can lie within tolerance everywhere and still shade incorrectly if it is locally rough. Localised feature loss is exactly the error mode a mean conceals, so a literature that claims feature preservation and measures it with symmetric Chamfer distance is using an instrument poorly matched to its own claim.

## 2.5 Industry context

Virtualized geometry systems have changed where reconstruction quality matters. Nanite (Karis, Stubbe and Wihlidal, 2021) builds a hierarchy of triangle clusters offline and selects a level at runtime, so source assets can carry detail far beyond what is rasterised in any frame. The quality ceiling therefore moves upstream, since a system that only decimates cannot restore a crease the reconstruction stage rounded off, and rebuilding by hand does not scale to the volumes these systems assume.

## 2.6 Summary and identification of the gap

The classical literature rounds sharp features because its smoothness prior is what makes reconstruction robust, and the extraction methods that recover features need normal information a raw scan does not reliably supply. Neural implicit methods replace the analytic prior with a learned one and have no structural obligation to smooth, but the same rounding reappears through spectral bias, answered in the field by changing the input encoding and by changing the activation function. The case for periodic activations is clear and widely cited, yet the comparisons supporting it vary architecture and training configuration together, so the difference cannot be attributed to the activation function specifically, and the reported metric is almost always a mean over the whole surface, insensitive by construction to the localised loss that feature preservation is about. The gap is twofold, an absence of comparisons in which the activation is the only variable, and an absence of evaluation sensitive enough to detect the failure the comparison is meant to resolve.

# 3. Methodology

## 3.1 Research design

This project uses a controlled comparative experiment. Two network architectures are trained to represent the same signed distance field under conditions held identical in every respect except the activation function, and the resulting reconstructions are compared across the metrics defined in Section 3.7.

An experimental design suits these research questions better than the alternatives considered. A purely analytical treatment would provide what each architecture can represent in principle but not what it does represent under a finite budget on a specific surface, which is what RQ1 asks. A benchmark across published methods would answer a different question, because published results differ in data, capacity, training length and extraction resolution simultaneously, and none isolates the activation function. Building both networks and controlling everything else is the only approach supporting a causal claim about that variable.

The design is deliberately narrow. One mesh, a set seed and a capacity setting are used. This supports statements about how these two architectures behave on this surface under this budget, and does not support general claims about SIREN and ReLU networks as classes. A difference of a few per cent between configurations cannot be distinguished from seed-to-seed variation in a single-run design, and no such distinction is claimed. Broadening the design to several meshes and seeds was considered and rejected on time grounds, since extraction alone costs several minutes per configuration. Narrowing the scope and reporting the limitation was preferred to broadening it and reporting each result unreliably. Section 5.6 develops these limits and Section 7.3 identifies the experiments that would address them.

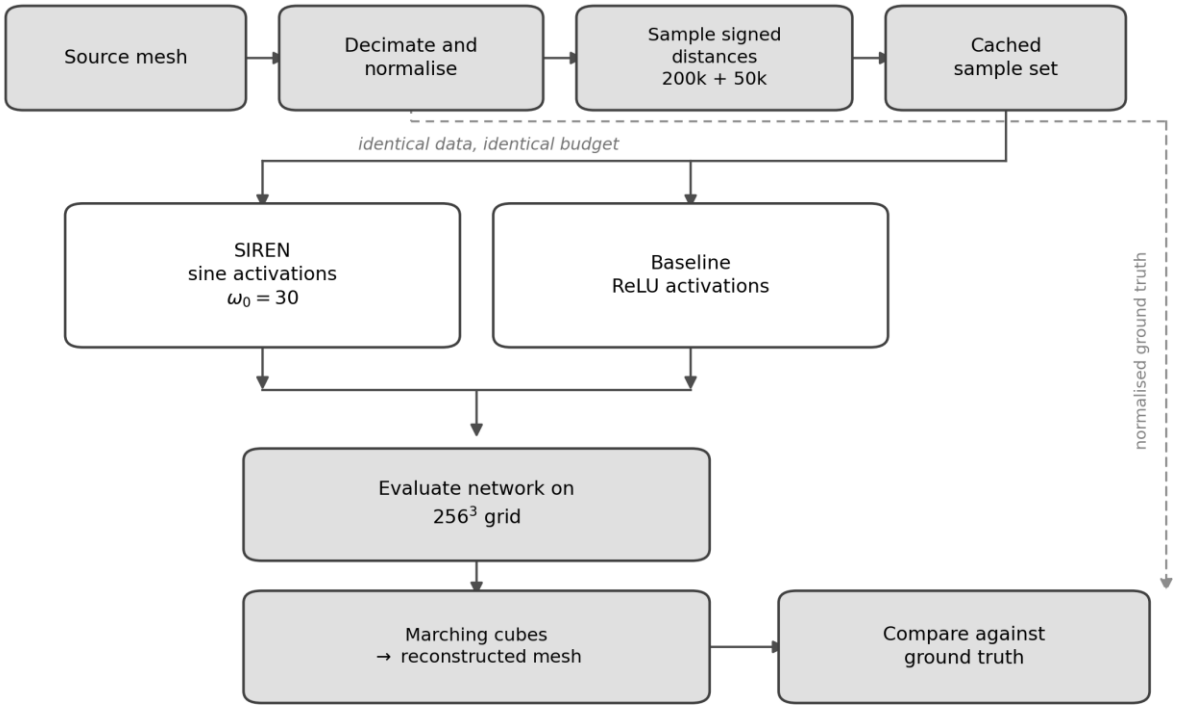
## 3.2 Pipeline overview

Figure 3.1 shows the processing pipeline. A triangle mesh is loaded, decimated and normalised. Points are sampled in and around the surface and paired with ground-truth signed distances to form the training set. A coordinate network is fitted to those pairs, then evaluated on a regular volumetric grid, and marching cubes extracts the zero-level set as a triangle mesh. That mesh is compared against the normalised ground truth to produce the metrics reported in Chapter 4.

Two properties matter for the comparison. Every stage before training is shared, because both architectures consume the same cached sample set, so training data is not a

variable. Every stage after training is shared and deterministic, so any difference in the output meshes originates in the fitted network rather than in extraction or evaluation.

**Figure 3.1** — Reconstruction and evaluation pipeline. Stages shaded grey are shared between both architectures. The dashed path indicates the normalised ground truth being carried forward for comparison. Both networks consume the identical cached sample set and receive identical training budgets.




## 3.3 Data and sample generation

The test object is the 'Male Base Mesh', submitted by nixor and credited to Paul Chen, obtained from Free3D under a Personal Use License (https://free3d.com/3d-model/male-base-mesh-6682.html). It is chosen because it combines a smooth surface that has a distinct continuous shape and includes sharp details that ultimately test each network's accuracy in the reconstruction of the mesh, namely the limbs and fingers.

The mesh is first decimated by quadric edge collapse to a target of 50000 faces. Decimation reduces the cost of the ground-truth distance queries, which dominate data preparation, without materially altering the surface at the scale the networks can represent: at the reported extraction resolution the grid spacing is 0.00784, considerably coarser than the edge length of the decimated mesh. The mesh is then checked to ensure it is watertight, otherwise it is repaired if necessary. Watertightness is not optional here: the sign of the distance field is

determined by an inside/outside test, which is only well defined for a closed surface. A mesh with boundary edges yields unreliable signs and a network trained on unreliable signs produces inaccurate and rough geometry at extraction. The final mesh is watertight with 24,461 vertices and 48,918 faces.

The mesh is then translated so its bounding box centroid lies at the origin and scaled so the furthest vertex's position is somewhere around the radius of 0.9, placing the whole object inside the [-1,1] cube with margin. All distances reported in this dissertation are in these normalised units.

Training data includes 250,000 point-distance pairs: 200,000 sampled near the surface and 50,000 sampled uniformly within the cube. The ratio is deliberately favoured towards the surface because the quantity of interest is the zero level set, and uniform sampling of the cube places most points far from any surface, where accuracy is irrelevant to the extracted mesh. The uniform component is retained because a network trained only on near-surface points has no constraint on its behaviour elsewhere and can produce inaccurate zero crossings in empty regions which marching cubes would extract as floating geometry.

The near-surface points are generated by sampling uniformly over the mesh surface and then displacing each point by an offset drawn uniformly and symmetrically about zero. Two displacement scales are used, half the points at +-0.01 per axis and half at +-0.05, so that the training set contains both fine details immediately adjacent to the surface and smoother structure at a scale comparable to the clamp bound. The displacement is symmetric so that points fall on both sides of the surface in roughly equal proportion, as an asymmetric displacement would bias the network's estimate of where the actual surface lies. Across the generated set, 35.2% of samples carry a negative distance.

Signed distances are computed by exact point-to-mesh query evaluated in chunks of 1,000 points to bound memory. The sign convention places negative values inside the surface. The complete sample set is cached to disk and loaded by both training scripts, so both architectures train on byte-identical data.

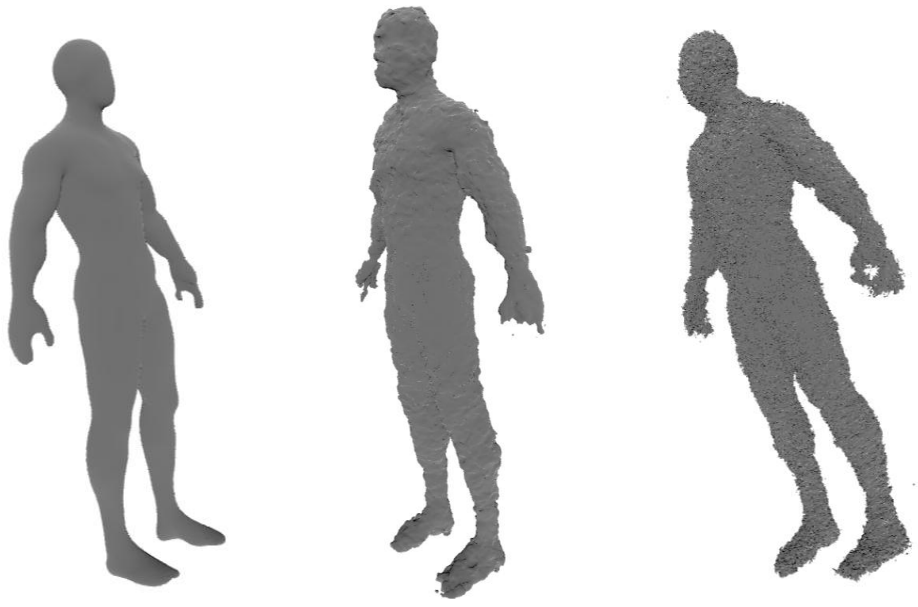
## 3.4 Network architectures

Both architectures are coordinate multilayer perceptrons mapping a 3D position to a scalar signed distance, with a hidden dimension of 256 across four hidden layers. This configuration was fixed in advance and applied identically to both networks. Wider and deeper SIREN variants were examined informally before the baseline was implemented (Appendix F), and the smallest configuration was retained. Because the research question targets the activation function specifically, holding the layer geometry constant is what allows any difference in reconstruction quality to be attributed to that variable. With no positional encoding the input

dimension is 3 in both cases and every layer shape matches, so the two networks contain exactly 198,657 parameters each. Therefore, capacity is not a candidate explanation for any observed difference.

The SIREN network uses sine activations throughout, following Sitzmann et al. (2020). The frequency scaling factor omega_0 is set to 30, the value reported in that work. This parameter multiplies the pre-activation of each layer and determines the spatial frequency band the network can represent. Values above 30 (45, 60) were tested during development and maintained the original shape, but made rougher, inaccurate and very noisy surfaces, rather than smooth, which is the opposite of the intended result. The published default was kept the same, supported by this observation as well as by the reference implementation. Figure 3.2 shows the comparison. SIREN also requires its own initialisation, the first layer drawing weights uniformly from ±1/in_features and hidden layers from ±√(6/in_features)/omega_0. This is not optional. The scheme exists to keep the distribution of activations stable through depth, and standard initialisation degrades the network substantially.

**Figure 3.2** — Reconstructions at omega_0 = 30, omega_0 = 45 and omega_0 = 60, under identical camera and lighting. The higher value recovers sharp edges more crisply but introduces high-frequency artefacts across regions that should be smooth.



The baseline replaces the sine activations with ReLU and is otherwise identical in width, depth, input encoding and output parameterisation. The initialisation necessarily differs, since SIREN's is derived from properties of the sine function and has no meaning for a piecewise-linear network. The baseline uses Kaiming normal initialisation (He et al., 2015) with the ReLU gain, drawing weights from a zero-mean normal of variance 2/fan_in and setting biases to zero. This is the standard choice for ReLU networks and the direct counterpart to SIREN's scheme, with both derived to ensure stable activation variance across network depth for their respective nonlinearities. Imposing a common initialisation would handicap whichever

network it was not designed for, so instead, using an appropriate scheme for each architecture should be a fair comparison.

In both networks the output layer is initialised to small magnitudes, so predictions begin close to zero across the domain. The schemes differ, SIREN drawing uniformly from ±√(6/hidden_dim)/omega_0 and the baseline from a zero-mean normal of standard deviation 0.1·√(2/hidden_dim), roughly three times wider. Under the loss formulation in Section 3.5 this affects only convergence speed. Under the superseded formulation described in Section 4.5, it was essential, since a network whose initial outputs fell outside the permitted range received no gradient.

No positional encoding is applied to either network, the number of frequency bands being zero in both cases. This is a deliberate restriction. Positional encoding is the established remedy for spectral bias in ReLU coordinate networks, so omitting it means the baseline represents an unaugmented ReLU MLP rather than the strongest available ReLU-based method. That limitation is revisited in Section 5.6.

## 3.5 Training configuration and the matched-budget control

Both networks are trained to regress the ground-truth signed distance directly, using an L1 loss between the predicted and target values. L1 is preferred to L2 because the sample distribution is concentrated near the surface, where a small number of geometrically difficult points would otherwise dominate a squared error term and pull the fit away from the bulk of the surface.

Target distances are truncated to ±0.1 before the loss is computed, following DeepSDF (Park et al., 2019). Truncation concentrates capacity around the zero-level set, which is the only region marching cubes reads. Distances beyond that band do not affect the extracted surface, and learning them accurately would consume capacity for no benefit. The cost is that neither network learns a valid distance field away from the surface, ruling out downstream uses such as sphere tracing, as discussed in Section 5.6.

The prediction is deliberately not truncated. Applying the same clamp to the prediction produces a loss with zero gradient wherever the prediction lies outside the bound, so a network whose outputs drift beyond ±0.1 receives no gradient and cannot recover. Section 4.5 reports the consequence of the earlier formulation, in which this occurred and was not visible in the loss curve.

Optimisation uses Adam (Kingma and Ba, 2015) with default momentum parameters, a batch size of 256 and 5,000 iterations. Batches are drawn by sampling indices uniformly with replacement rather than by iterating over epochs, so the 5,000 iterations correspond to 1,280,000 sample draws against a pool of 250,000 points, or 5.12 effective epochs. Both

architectures receive an identical number of gradient updates on identical batches, which is what makes the compute budget comparable. Training was performed on CPU and takes between 56 and 72 seconds per configuration.

The learning rate is fixed at 1e-4 for both networks. This value was not chosen by preference but forced by the behaviour of SIREN at higher rates. A rate of 1e-3 is conventional for a ReLU MLP of this size and the baseline trains without difficulty at it, but SIREN fails to converge. Its training loss falls from 0.06023 to 0.03288 over the first 500 iterations and then oscillates between 0.030 and 0.035 for the remaining 4,500, ending at 0.03009 against 0.00258 at the lower rate. The field it produces is 23.6% negative by volume, compared with 0.9% for the converged network, and surface extraction from it exhausted available memory. A rate of 1e-3 therefore cannot serve as a common setting, since it would compare a converged baseline against a network that never fitted the data. A plausible explanation lies in the scale of the weights rather than of the gradients. Adam normalises each update by a running estimate of gradient magnitude, so each parameter moves by up to roughly the learning rate regardless of how large its gradient is. SIREN's hidden weights are initialised within $\pm\sqrt{(6/256)}/30 \approx \pm 0.0051$, whereas the baseline's Kaiming weights have a standard deviation of $\sqrt{(2/256)} \approx 0.088$. A step of 1e-3 can therefore change a typical SIREN hidden weight by around 40%, against roughly 1.4% for the baseline, and the omega_0 factor carries that change directly into every sine pre-activation. This is offered as a hypothesis consistent with the observation rather than a demonstrated mechanism, as update magnitudes were not instrumented. Practically it indicates that SIREN requires a substantially lower learning rate than a ReLU network of the same size, a cost that a comparison at a single tuned rate does not capture.

No learning-rate schedule, weight decay, gradient clipping or early stopping is applied. Each would introduce a further hyperparameter whose optimal value might differ between architectures, reintroducing the confound the matched-budget design exists to remove. Training runs for the full 5,000 iterations and the final weights are used, with no selection against a validation set.

All runs use a fixed seed of 0 for both NumPy and PyTorch, and both architectures load the same cached sample set, so the two networks see identical data. Results come from a single run per configuration. No repeats across seeds were performed, and the variance attributable to initialisation is unmeasured. Section 5.6 returns to this.

## 3.6 Surface extraction

The trained network defines an implicit surface as the set of points where its output is zero. Recovering an explicit mesh requires evaluating the network on a regular grid and extracting the zero level set, for which marching cubes (Lorensen and Cline, 1987) is used via its scikit-

image implementation. The network is evaluated at every point of an $R^3$ grid spanning [-1, 1], in chunks to bound memory, and the resulting scalar volume is passed to marching cubes at level zero. R = 64 was used during development, where 262,144 evaluations complete in seconds and permit rapid iteration. All reported results use R = 256, giving 16,777,216 evaluations and a grid spacing of 0.00784.

That spacing sets a ceiling on what any reconstruction here can express, independent of the network. Features thinner than roughly one grid cell cannot appear in the extracted mesh even if the network has learned them exactly, and features spanning two or three cells are recovered with visibly reduced sharpness. The fingers and toes of the test mesh approach this limit, so a region left uncovered in Section 4.3 may reflect the extraction resolution rather than the network. Extraction is common to both architectures and does not bias the comparison, but it bounds the absolute quality either can reach.

Extraction is also the dominant cost, accounting for between three and five times the cost of training in every configuration reported in Section 4.4, because it requires more forward passes than the entire training run.

A check is applied before extraction. If the sampled volume contains no sign change then no surface exists, and extraction is aborted with an explicit error. This was added after a training defect produced a field of uniform sign, which the extraction library reported only as an out-of-range level value (Section 4.5).

## 3.7 Evaluation protocol

Reconstruction quality is assessed by comparing the extracted mesh against the ground truth after identical normalisation, so all distances are in normalised model units in which the ground truth has a bounding radius of 0.9. Both surfaces are sampled at 200,000 points using a fixed seed applied identically to every configuration, so differences between results reflect the reconstructions rather than the sampling.

Two directed distances underlie most of the metrics. Accuracy is the distance from each reconstructed point to the nearest ground-truth point and measures the extent to which the network has invented geometry that does not exist. Completeness is the distance from each ground-truth point to the nearest reconstructed point and measures the extent to which real geometry has been missed. The distinction matters because the two failure modes are not interchangeable. A reconstruction that omits a thin feature and one that adds a spurious blob are both wrong, but for different reasons and with different downstream consequences.

Symmetric Chamfer-L1 is the sum of the two directed means. It is included because it is the standard aggregate in the literature and is required for comparability with prior work, but it is a lossy summary. Because it sums the two directed errors, a configuration worse on accuracy

and better on completeness can produce the same value as one with the opposite profile. It is reported alongside, not instead of, the distributional measures below.

Both directed distances are also reported at the 50th, 90th and 99th percentiles and at the maximum. The mean is dominated by whichever tail is heavier, and two distributions with the same mean can differ substantially in shape. The median describes the typical point on the surface, while the 99th percentile and maximum describe the worst regions. For work concerned with feature preservation the tail is the quantity of interest, since a reconstruction is judged unusable by its worst region rather than its average one. The ratio of mean to median is reported as a compact measure of skew.

The uncovered fraction is the fraction of ground-truth samples with no reconstructed point within a tolerance tau, evaluated at ten values from 0.002 to 0.030. A single threshold is not sufficient, because the choice of tau is arbitrary and a ranking obtained at one value may not hold at another. Reporting the whole curve makes that dependence visible rather than hiding it inside a choice made by the author. The range brackets the $256^3$ grid spacing of 0.00784. F-score at each tau is reported alongside, combining recall (one minus the uncovered fraction) with precision, the fraction of reconstructed points within tau of the ground truth.

Normal consistency is the mean absolute dot product between the normal at each reconstructed sample and the normal at its nearest ground-truth point. It is included because the distance metrics are insensitive to a failure mode that matters visually. A surface can lie within tolerance everywhere while still being locally rough, and such a surface shades incorrectly under any lighting model using normals. The absolute value is taken so a consistently inverted normal is not penalised, since orientation conventions differ between extraction implementations.

All metrics come from a single run per configuration, so variance attributable to initialisation and batch sampling is not measured. Small differences between configurations should be read with that in mind. Section 5.6 returns to this.

## 3.8 Experimental procedure and reproducibility

All the experiments were run in Google Colab on CPU. The implementation uses PyTorch for the networks, trimesh for mesh handling and ground-truth distance queries, scikit-image for marching cubes, and SciPy for the nearest-neighbour queries used in evaluation. Version numbers are given in Appendix E.

Random number generators of NumPy and PyTorch are seeded with 0 at the start of every run, before any sampling or network construction. The generated sample set is cached to disk on first creation and loaded afterwards, so all configurations train on identical data regardless

of execution order. Metric computation uses its own fixed seed for surface sampling, applied identically to every mesh.

Each configuration is a single run of one script with one changed value. The full hyperparameter set is given in Appendix C, the values that differ between configurations are the activation function and the learning rate, and nothing else.

On the reported hardware, ground-truth sampling is a one-time cost of approximately six minutes. Per-configuration costs include 56–72 seconds for training and 224–290 seconds for surface extraction, while metric computation for all three configurations combined takes under a minute.

Reproducing the reported numbers requires the source mesh, the two training cells, the evaluation cell and the failure-analysis cell, all submitted with this dissertation, together with the package versions in Appendix E. Running the SIREN script first regenerates and caches the sample set, the remaining configurations, and load it. Exact reproduction of the floating-point values additionally requires the same library versions, since the mesh decimation and sampling routines are not guaranteed stable across releases.

## 3.9 Ethical and professional considerations

This project involves no human or animal participants, no personal data and no user study, and therefore raises none of the concerns arising in research with human subjects. Ethical approval for this project was confirmed by the module leader, Dr Malekmohamadi, and the confirmation is reproduced in Appendix B.

The single data item used is a triangle mesh obtained from Free3D under a Personal Use License, as detailed in Section 3.3, and that licence permits academic use of this kind. No proprietary or restricted assets were used, and no third-party code was incorporated beyond the open-source libraries listed in Section 3.8, each used within its licence terms. The computational cost of the work is modest, all experiments having run on shared CPU resources over hours rather than days. This is noted rather than claimed as significant, because energy cost is a live concern in machine learning research and disclosing it is straightforward.

The professional consideration bearing most directly on this project is honest reporting. The BCS Code of Conduct requires members to avoid misrepresenting their work, and two aspects of the results are relevant. First, the headline result is not a clean win for either architecture, the two best configurations being indistinguishable on the standard aggregate metric and reporting a difference where none is demonstrated would misrepresent the finding. Second, several defects were identified in the experimental code during the project, and their correction changed the direction of the reported result. These are documented in Section 4.5

rather than omitted. Reporting the process by which a result was reached, including the errors, is a condition of that result being usable by anyone else.

# 4. Results

## 4.1 Quantitative comparison

Three configurations are reported: SIREN and the ReLU baseline at the matched learning rate of 1e-4, and the baseline at 1e-3, the better of the two rates tested for it. All use identical training samples and seeds, 5,000 iterations at a batch size of 256, and marching cubes at $256^3$ over [-1, 1], a grid spacing of 0.00784. Metrics derive from 200,000 seeded surface samples per mesh. All distances are in normalised model units, in which the ground-truth mesh has a bounding radius of 0.9. The extracted meshes are comparable in complexity with 42,510, 45,162 and 43,284 vertices, so the differences below are not attributable to tessellation density.

All three networks converge. Final training L1 is 0.00258 for SIREN, 0.00394 for the baseline at 1e-4, and 0.00275 for the baseline at 1e-3.

Table 4.1 reports the full metric set. Symmetric Chamfer-L1 is 0.008395 for SIREN, 0.016943 for the baseline at the matched rate, and 0.008402 for the baseline at 1e-3. The first two differ by a factor of 2.0; the first and third differ by 0.000007, or 0.08%.

The percentile breakdown separates the two configurations that the aggregate does not. At the median the baseline at 1e-3 records the lower accuracy error, 0.002775 against SIREN's 0.003130, and the lower completeness error, 0.002805 against 0.003208. At the 99th percentile the accuracy ordering reverses: SIREN records 0.012050 against 0.039787, with maxima of 0.018979 and 0.059760, equivalent to 2.1% and 6.6% of the bounding radius. Completeness at the 99th percentile does not reverse, the baseline recording 0.026629 against SIREN's 0.040389, with maxima of 0.055729 and 0.099631.

The ratio of mean to median accuracy error is 1.20 for SIREN, 1.55 for the baseline at 1e-3, and 3.62 for the baseline at 1e-4. For completeness the corresponding ratios are 1.44, 1.47 and 1.89.

The baseline at the matched rate reaches a maximum accuracy error of 0.141698, 15.7% of the bounding radius, while its median accuracy error of 0.003097 is within 1.1% of SIREN's.

Mean absolute normal consistency is 0.9812 for SIREN, 0.9620 for the baseline at 1e-3, and 0.8881 at 1e-4.

**Table 4.1** — Reconstruction quality for the three configurations. All distances are in normalised model units, in which the ground-truth mesh has a bounding radius of 0.9. Distance metrics are computed from 200,000 seeded surface samples per mesh. Final training L1 is the loss at iteration 4,999, computed on clamped targets.

| METRIC | SIREN 1e-4 | ReLU 1e-4 | ReLU 1e-3 |
|---|---|---|---|
| Parameters | 198,657 | 198,657 | 198,657 |
| Final training L1 | 0.00258 | 0.00394 | 0.00275 |
| Symmetric Chamfer – L1 | 0.008395 | 0.016943 | 0.008402 |
| Normal Consistency | 0.9812 | 0.8881 | 0.9620 |
| Accuracy – mean | 0.003765 | 0.011202 | 0.004288 |
| Accuracy – p50 | 0.003130 | 0.003097 | 0.002775 |
| Accuracy – p90 | 0.006949 | 0.023261 | 0.006782 |
| Accuracy – p99 | 0.012050 | 0.121113 | 0.039787 |
| Accuracy – max | 0.018979 | 0.141698 | 0.059760 |
| Completeness – mean | 0.004630 | 0.005741 | 0.004115 |
| Completeness – p50 | 0.003208 | 0.003035 | 0.002805 |
| Completeness – p90 | 0.007729 | 0.012284 | 0.007380 |
| Completeness – p99 | 0.040389 | 0.047501 | 0.026629 |
| Completeness – max | 0.099631 | 0.099202 | 0.055729 |
| Vertices | 42,510 | 45,162 | 43,284 |
| Faces | 85,016 | 90,084 | 86,564 |

## 4.2 Threshold sensitivity

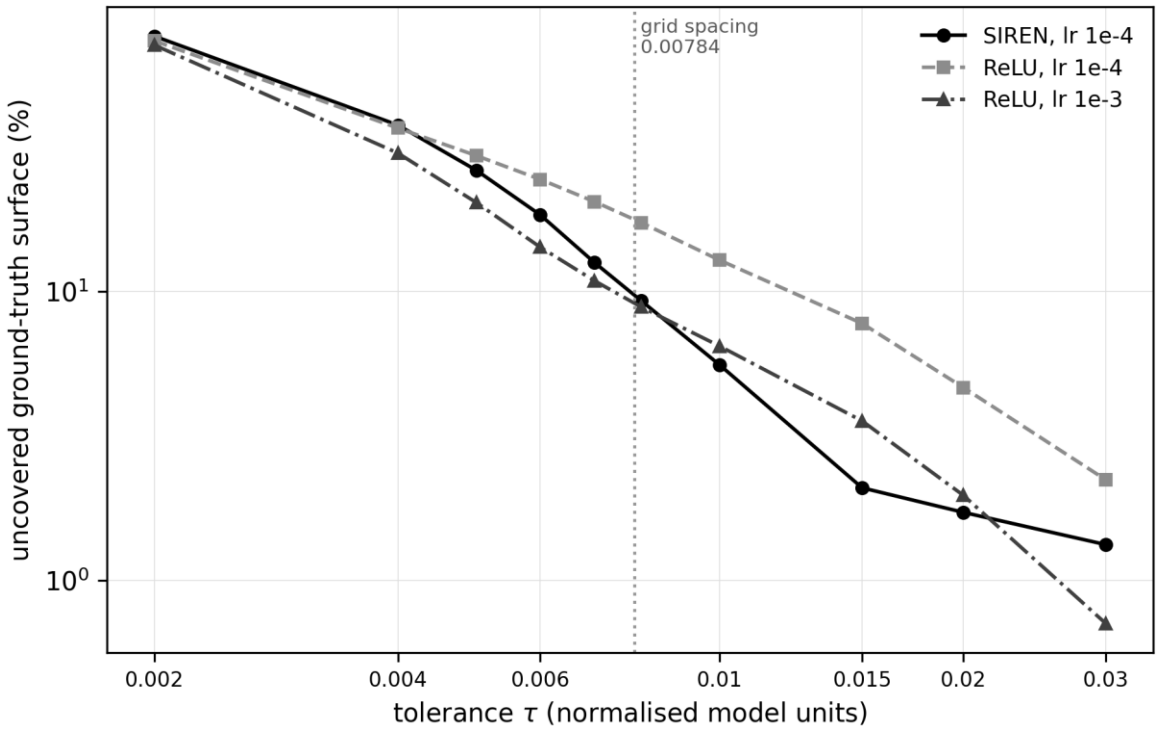
Figure 4.1 plots uncovered ground-truth surface fraction against tolerance τ, for τ from 0.002 to 0.030 on a logarithmic vertical axis, with the $256^3$ grid spacing of 0.00784 marked.

The ordering between SIREN and the baseline at 1e-3 reverses twice. Below the grid spacing the baseline leaves less surface uncovered: 20.29% against 26.24% at τ = 0.005, 14.21% against 18.38% at τ = 0.006, and 8.82% against 9.27% at τ = 0.008. Between τ = 0.008 and τ = 0.010 the curves cross, and SIREN leads at τ = 0.010 (5.56% against 6.46%), τ = 0.015 (2.09% against 3.56%) and τ = 0.020 (1.72% against 1.97%). Between τ = 0.020 and τ = 0.030 they cross a second time, the baseline recording 0.71% against SIREN's 1.33% at τ = 0.030.

The ordering against the baseline at 1e-4 also reverses. That configuration leaves less surface uncovered than SIREN at τ = 0.002 (73.56% against 76.34%) and τ = 0.004 (36.71% against 37.60%), and more at every larger threshold, from 29.48% against 26.24% at τ = 0.005 to 2.23% against 1.33% at τ = 0.030.

F-scores cross once. The baseline at 1e-3 is higher up to τ = 0.007 (0.8982 against 0.8880) and SIREN is higher from τ = 0.008 (0.9211 against 0.9177) through τ = 0.030 (0.9933 against 0.9875).

**Figure 4.1** — Fraction of the ground-truth surface left uncovered as a function of tolerance τ, for the three configurations. Both axes are logarithmic. The dotted vertical line marks the $256^3$ marching cubes grid spacing of 0.00784. Tolerances are in normalised model units, in which the ground-truth mesh has a bounding radius of 0.9. A point is counted as uncovered if no point of the reconstructed surface lies within τ of it; values are computed from 200,000 seeded ground-truth surface samples.




## 4.3 Failure region analysis

Figure 4.2 shows the ground-truth surface with distance to each reconstruction mapped to colour on a shared scale of 0 to 0.030, under identical camera and lighting.

At τ = 0.010, 3,192 of the 24,461 ground-truth vertices lie beyond the threshold for SIREN, 5,494 for the baseline at 1e-4, and 3,433 for the baseline at 1e-3. These sets are not distributed across the surface: they form 38, 39 and 35 connected components respectively. The largest single component spans 1,016, 1,153 and 659 vertices, and the five largest account for 57.8%, 65.1% and 59.1% of all uncovered vertices in each case. Mean component size is 84, 141 and 98 vertices. Failures are therefore contiguous regions of surface rather than distributed noise, in every configuration.

The number of failure regions is close to constant across architectures and learning rates, varying only between 35 and 39, while their mean size varies by a factor of 1.7 between the best and worst configuration.

The uncovered regions occupy parts of the surface with substantially higher curvature than the remainder. Mean absolute discrete curvature over uncovered vertices is 0.2125 for SIREN, 0.1932 for the baseline at 1e-4 and 0.2191 for the baseline at 1e-3, against 0.0984, 0.0901 and 0.0960 over the covered vertices, giving ratios of 2.16×, 2.14× and 2.28×. The ratio varies by less than 7% across the three configurations.

Visually the affected regions coincide across all three configurations, concentrating on thin, high-curvature extremities: the fingers, the toes and the outer ears. The torso, the limb surfaces and the back, large regions of low curvature, are reconstructed to within the threshold in every configuration.

The architectures differ in the severity of these failures rather than their location. SIREN recovers the feet and toes substantially better than either baseline configuration, where the toes are lost almost entirely; the baseline at 1e-4 additionally shows a large failure region at the neck and shoulder junction that is absent from the other two. SIREN in turn exhibits mild elevated error distributed across the hip and thigh, regions where both baseline configurations remain within threshold.

**Figure 4.2** — Distance from the ground-truth surface to each reconstruction, mapped to vertex colour. Blue denotes zero distance, red denotes 0.030 normalised model units or greater. Colour scale and camera are identical within each row. Left to right in each row: SIREN at 1e-4, ReLU baseline at 1e-4, ReLU baseline at 1e-3.

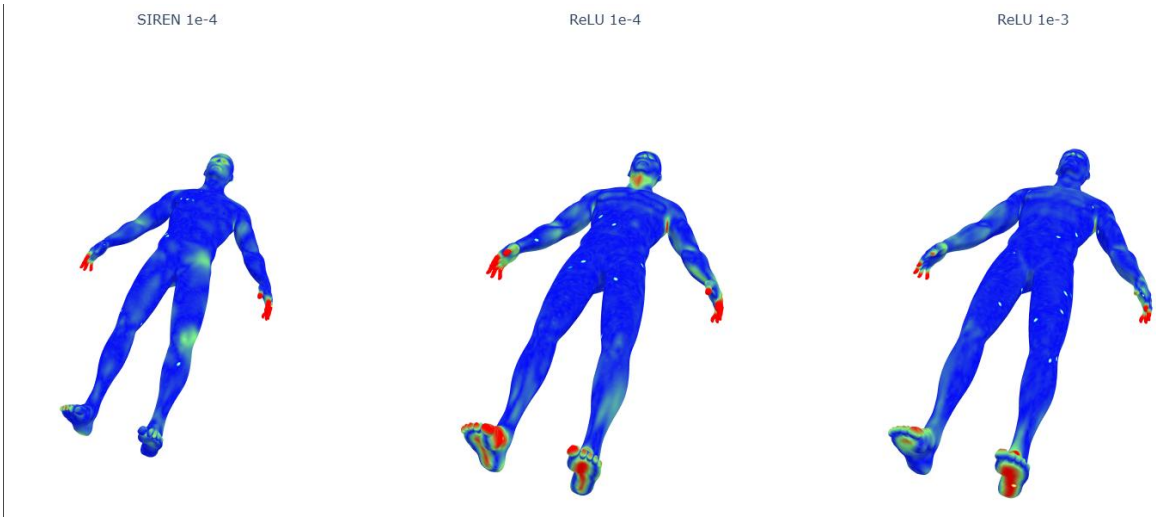



**Figure 4.2(a)** — Three quarter view




**Figure 4.2(b)** — Front view

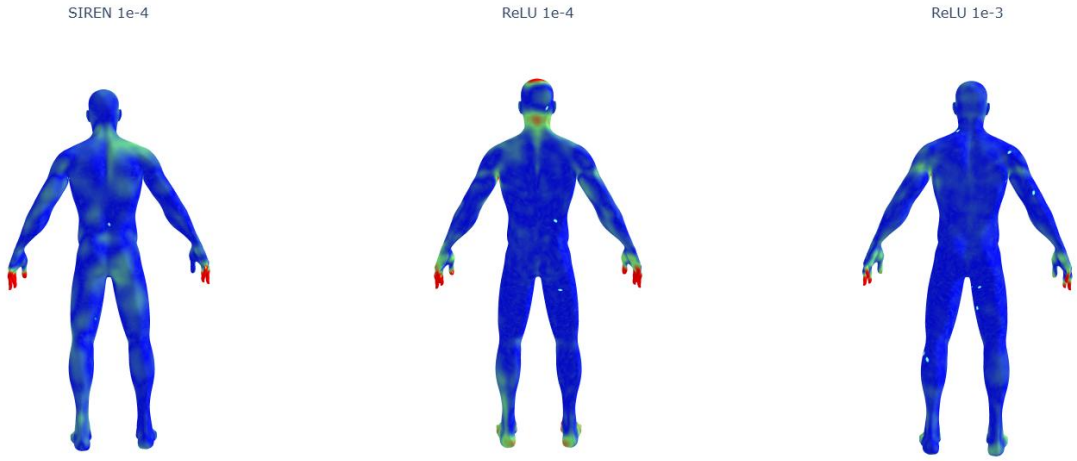



**Figure 4.2(c)** — Rear View

Note that the uncovered fractions in this section are computed over mesh vertices and are not directly comparable with the area-weighted figures of Section 4.2, because vertex density is higher in regions of high curvature, so vertex-based fractions are correspondingly higher.

## 4.4 Computational cost

Ground-truth SDF sampling is performed once and cached, taking approximately 6 minutes on first generation and shared across all configurations, and is excluded from the per-architecture figures below. Both networks contain exactly 198,657 parameters. All runs were executed on CPU.

Training for 5,000 iterations takes 63.6 s for SIREN, 56.2 s for the baseline at 1e-4 and 71.6 s for the baseline at 1e-3. Surface extraction at $256^3$ takes 289.8 s, 227.7 s and 224.1 s

respectively, comprising 16,777,216 network evaluations followed by marching cubes. Extraction therefore accounts for between 3.1 and 4.6 times the cost of training in every configuration. Peak memory was not instrumented, as the measurement available applies to CUDA allocations only and all runs were CPU-bound.

## 4.5 Superseded runs and corrections

Three defects were identified over the course of the experiments, each of which invalidated the results obtained before it was corrected. They are reported here rather than omitted.

First, the two implementations sampled training points independently and without a shared seed, and the baseline's near-surface perturbation was parameterised such that samples were displaced inward by a systematic offset rather than distributed symmetrically about the surface (Section 3.3). The two networks therefore trained on different data, and the baseline trained on data that misrepresented the region of interest. Second, the baseline reserved 5% of samples as a held-out set while SIREN did not, so the two networks trained on 237,500 and 250,000 points respectively. Third, and most consequentially, the training loop clamped the network's prediction as well as the target before computing the L1 loss. Because the clamp operation has zero gradient outside its bounds, any network whose outputs drifted beyond ±0.1 ceased to receive gradient signal entirely. This condition was not visible in the loss curve, which continued to report a plausible constant value, and was detected only when surface extraction failed on a field containing no zero crossing.

Under the earliest configuration, with mismatched learning rates and independent sampling, the baseline led on symmetric Chamfer-L1. The figures from that run were not retained, as the defects above invalidate it in three separate respects, it is noted here only because it produced the opposite ordering to the results.

All three defects were corrected and every configuration retrained. No result in Chapter 5 rests on a superseded run. They are reported because the direction of the headline result changed during correction, and because the third defect was undetectable from the training loss alone.

# 5. Discussion and Critical Evaluation

## 5.1 RQ1: does SIREN preserve sharp features better?

Under the matched budget comparison defined in Section 3.5 the answer is yes. At a common learning rate of 1e-4 the SIREN network is better than the ReLU baseline on every metric in Table 4.1 except the two medians and maximum completeness, where the baseline is lower by 1.1%, 5.7% and 0.4% respectively. Under the broader question a practitioner would ask, whether SIREN beats a ReLU MLP at the better of the two rates tested, the answer is conditional. SIREN preserves thin details and surface orientation better, but it is no more accurate at the typical point on the surface, and the advantage is bought with a much narrower tolerance for optimiser settings.

Taking the matched comparison first, symmetric Chamfer L1 favours SIREN by a factor of 2.0, at 0.008395 against 0.016943, and normal consistency by 0.9812 against 0.8881. At the median the two are almost identical, the baseline in fact recording marginally the lower error at 0.003097 against 0.003130. By the 99th percentile the ordering reverses by a factor of 10.1, and the baseline reaches a maximum accuracy error of 15.7% of the bounding radius against SIREN's 2.1%. The ReLU network fits most of the surface as well as SIREN does and fails badly on a small part of it.

Matching the learning rate influences the comparison. Section 3.5 fixed it at 1e-4 because SIREN does not converge at 1e-3, not because 1e-4 suits both architectures. Matching the rate equalises the nominal step size while leaving each architecture at a different distance from its own best condition, which matches the number rather than what the number does. It is a trade-off between competing factors rather than the removal of one, since tuning separately mixes activation effects with tuning effort, while a common rate makes the comparison dependent on distance from the best performing setting. Reporting both configurations, as Chapter 4 does, is the only honest response available in a single run design.

Against the baseline at 1e-3 the result is genuinely mixed. Symmetric Chamfer L1 is 0.008395 against 0.008402, a difference of 0.08% that one seed cannot distinguish from noise and which should be read as a tie. SIREN leads in the accuracy tail, 0.012050 against 0.039787 at the 99th percentile and 0.018979 against 0.059760 at the maximum, and on normal consistency at 0.9812 against 0.9620. The baseline leads at the accuracy median, at the 90th percentile, and at every reported percentile of completeness, including 0.026629 against 0.040389 at the 99th. SIREN's accuracy advantage opens only between the 90th and 99th percentiles, making it a property of the worst tenth of the reconstructed surface and of nothing else.

The two architectures are therefore wrong in opposite directions. SIREN invents very little geometry that is not there, nothing it produces straying more than 2.1% of the bounding

radius from the ground truth, and its main weakness is the extent of what it leaves out. It leaves less surface uncovered than the baseline between tolerances of roughly 0.009 and 0.020, but the regions it does miss it misses further, which drives both its heavier completeness tail and the second curve crossing at 0.030 in Section 4.2. The answer to RQ1 is thus a qualified yes. SIREN preserves thin features of high curvature and surface orientation substantially better, but it holds no advantage at the typical point, performs worse for the least complete cases, and its failure to converge at a rate unremarkable for a ReLU MLP is a cost a comparison at a single tuned setting would not have exposed.

## 5.2 RQ2: failure modes and what they imply

The most informative result in Section 4.3 is a consistency rather than a difference. The uncovered vertices at a tolerance of 0.010 form 38, 39 and 35 connected components across the three configurations, the five largest account for between 57.8% and 65.1% of uncovered vertices in each case, and the ratio of curvature over uncovered vertices to curvature over covered vertices is 2.16, 2.14 and 2.28, a spread of under 7%. The affected regions appear in the same locations: the fingers, the toes, and the outer ears. Only the degree of degradation varies, with mean component sizes of 84, 141, and 98 vertices. Both architectures struggle with the same types of geometry, while the activation function determines the extent of the effect rather than its location.

An explanation must therefore lie in what the configurations share: the mesh, the sample set, the extraction grid and the evaluation. Four mechanisms are plausible. The grid spacing of 0.00784 is only a little finer than the fingers and toes, so a feature spanning two or three cells is poorly resolved regardless of how accurately the network represents it. Surface points are drawn uniformly by area, so the extremal regions receive samples in proportion to area rather than complexity, and the regions requiring the most detail receive the fewest. The value of omega_0 bounds the frequencies SIREN can express, and spectral bias bounds the baseline, both predicting failure concentrated at high curvature. Finally, wherever two sheets of surface lie close together, between adjacent fingers or between ear and head, the signed distance field contains a ridge along the medial axis between them, where its gradient reverses direction. The training labels there are correct, since distances come from an exact query against the whole mesh, but the network must represent a crease in the field even where the surface on either side is smooth, and spectral bias applies to that crease as much as to one on the surface.

The data cannot separate these, since all three configurations share one mesh, one sampling scheme and one extraction resolution. It does establish that the grid ceiling is not the whole story, because if extraction were the binding constraint everywhere all three would fail identically in those regions, and they do not, SIREN recovering the feet and toes visibly better while mean component size varies by a factor of 1.7. The pipeline appears to set a floor and

the architecture to determine how close a configuration lands to it. The medial-axis explanation is hardest to rule out, because the narrow regions of this mesh are both highly curved and close to other parts of the surface, making these two explanations difficult to distinguish in the test object.

The implication is uncomfortable and should be stated plainly. If sampling and extraction fix where the pipeline fails while the activation function moves only how badly, then the practical lever for feature preservation here is the data and extraction stages rather than the choice of nonlinearity. The comparison remains informative, since a reduction of 3.3 times in the accuracy tail is worth having, but it places the activation function as a second order effect inside a pipeline whose first order limits lie elsewhere.

## 5.3 RQ3: what the metric divergence means

The most consequential result in this dissertation is that symmetric Chamfer L1 separates SIREN and the baseline at 1e-3 by 0.08% while the two reconstructions are not alike. A study reporting only the overall result would have called the activation function irrelevant here, and no care in running the experiment would have caught the error.

The mechanism is arithmetic. Symmetric Chamfer L1 is the sum of the two directed means. For SIREN that sum is 0.003765 plus 0.004630, giving 0.008395, and for the baseline at 1e-3 it is 0.004288 plus 0.004115, giving 0.008402. SIREN's accuracy mean is 12.2% lower than the baseline's and its completeness mean is 12.5% higher, and the two displacements very nearly cancel. The near equality of the overall score is not evidence that the reconstructions agree, but rather that they differ in opposite directions by almost the same amount. Nothing in the metric prevents this and nothing in the reported figure reveals it, yet reporting the directed means separately costs nothing, since the symmetric value is their sum.

A second and more familiar limitation adds to the first. A mean over 200,000 samples is dominated by the samples that are fine. The baseline at 1e-4 records a median accuracy error within 1.1% of SIREN's and a maximum 7.5 times larger, its ratio of mean to median being 3.62 against 1.20. A mean answers how good a surface is on average, not whether an asset is usable, because an asset is rejected on its worst region rather than its typical one.

The threshold-based metrics show a third failure, that the ranking depends on a parameter the author chooses. The uncovered fraction curves for SIREN and the baseline at 1e-3 cross twice, near 0.009 and again between 0.020 and 0.030, and the F score curves cross once near 0.0075. At 0.005 the baseline leads by six percentage points, at 0.015 SIREN leads by 1.5, and at 0.030 the baseline leads again, so any single tolerance report would be defensible and arbitrary at the same time. Reporting a point hides a choice made by the author inside a figure the reader will take for a measurement.

A practitioner should therefore choose the metric that matches the failure carrying cost downstream. Where reconstructions feed an authoring pipeline in which the worst regions are repaired by hand, the relevant quantities are the directed tails and the uncovered fraction at the tolerance the next stage enforces, and on those SIREN is the better choice on this surface. Where the output serves as a collision approximation or a lower level of detail, the overall measure is sufficient, and the two architectures perform similarly at this budget.

The conclusion for the field is narrower but firmer. Work claiming feature preservation on the evidence of one symmetric Chamfer figure reports a statistic structurally insensitive to the failure mode it claims to address, and these results show that the insensitivity can arise by cancellation rather than coincidence. A minimum standard follows and costs almost nothing. Report the two directed means rather than only their sum, report at least one tail percentile beside the mean, and report coverage across a range of tolerances with the grid spacing marked, so a reader can see which part of the range is bounded by the network and which by the isosurfacing.

## 5.4 Relation to previous work

The matched budget result agrees with the qualitative claim in Sitzmann et al. (2020), the difference in accuracy at the 99th percentile expressing quantitatively the sharpness that work demonstrates on renders. The tension is over scope rather than direction. Against a baseline at the better of the two learning rates, the advantage contracts to the accuracy tail and to normal consistency, vanishes at the median and reverses in completeness. Part of what is reported as a representational advantage of periodic activations is, on this evidence, an advantage in optimisation behaviour at a shared setting. The metrics on which SIREN retains an advantage are also the ones a render exposes, since a tail failure is a visible artefact while a 12% difference in median error is not.

The optimiser sensitivity in Section 3.5 is less a contradiction of that work than a consequence of its parameterisation, and is consistent with the rate of 1e-4 used in its own experiments. The initialisation proposed by Sitzmann et al. divides hidden weights by omega_0 so that activations stay stable through depth, which also makes those weights more than an order of magnitude smaller than their ReLU counterparts, while Adam sets each parameter's step by the learning rate rather than by the weight's scale. A learning rate matched in parameter units is therefore not matched in its effect on the function. This would explain both the divergence of SIREN at 1e-3 and the higher final loss of the baseline at 1e-4, and it reinforces the argument in Section 5.1 that the matched rate equalises the number rather than what the number does. Update magnitudes were not measured, so the observation stands independently of this explanation.

The comparison should also be positioned against NGLOD and Instant NGP, which obtain capacity for high frequency detail from the input representation rather than the activation function. The baseline here carries no positional encoding, so this project compares SIREN against the weakest reasonable ReLU network rather than the standard remedy for spectral bias. That the unencoded baseline still ties SIREN on the aggregate at the better of its two tested rates is consistent with the movement of the field toward encodings, though encoding was not tested and the suggestion is not evidence.

## 5.5 Critical evaluation of the artefact and of the objectives

Assessed against the objectives in Section 1.3, the pipeline objective is met, with the system carrying a triangle mesh through sampling, training, extraction and evaluation without manual input. The matched configuration objective is met in the sense specified, through identical cached samples, parameter counts of 198,657, iteration counts and extraction, and only partly met in a stricter sense, since Section 5.1 argues that matching the learning rate is itself a substantive choice. The evaluation protocol objective is met and is the strongest part of the artefact. The comparison objective is met, Chapter 4 reporting the full metric set for all three configurations, and the disagreement between aggregate and tail metrics that it exposed is the subject of Section 5.3. The failure analysis objective is met descriptively but only partly analytically, since Section 5.2 shows the experiment cannot discriminate between the mechanisms explaining the pattern Section 4.3 establishes.

The software performs most reliably where determinism was deliberately incorporated into the design. Caching the sample set and loading it from both training scripts removes training data as a variable, and the shared deterministic path after training means any difference in the output meshes originates in the fitted network.

The weaknesses are structural and were expensive. Maintaining two parallel training scripts rather than one entry point parameterised by activation is the direct cause of two of the three defects in Section 4.5, since duplicated code diverges silently and a holdout present in only one script is exactly what duplication produces. The only training diagnostic was the loss curve, which was uninformative in precisely the failure it most needed to catch, a network receiving no gradient reporting a plausible constant loss for 4,500 iterations. Three cheap instruments would have caught all three defects, an assertion on the mean displacement of near surface samples, a logged fraction of predictions outside the clamp band, and a logged gradient norm.

## 5.6 Limitations

The most important limitation is the test object. The results rest on one mesh and that mesh is organic. Chapter 1 motivates the problem with the creases and corners of mechanical parts, while the geometry that stresses the reconstruction here is thin extremities of high curvature, the fingers, the toes and the ears. These are legitimately sharp features, but they are a different geometry class from the CAD surfaces the introduction invokes, and no claim about crease or corner preservation is supported here.

All results derive from a single seed with no repeats, so no variance is measured and no interval can be attached to any figure in Table 4.1. Differences of a few per cent are uninterpretable, explicitly including the 0.08% agreement in symmetric Chamfer L1 and the 12.8% gap in median accuracy error. The medians are further compressed by the evaluation itself, since directed distances are measured between two independent 200,000-point samplings rather than to the surface. Even a perfect reconstruction would therefore record a median of roughly 0.0014 and leave about a fifth of the surface uncovered at $\tau = 0.002$. The effect is negligible from $\tau \approx 0.004$ upward, so the tail measures and curve crossings are unaffected. The argument rests only on the large ratios reported above.

The comparison is also narrow on both sides. Extraction resolution places a ceiling on absolute quality that is nowhere separated from network behaviour, since at a grid spacing of 0.00784 the features that fail are a few cells wide, and one run at $512^3$ would have separated the two causes. There is no classical baseline, so nothing anchors these results to screened Poisson reconstruction. The ReLU network carries no positional encoding and is therefore not the strongest available baseline, and no hyperparameter was swept, omega_0 = 30 having been fixed on an informal pilot rather than a measured sweep.

Several narrower limitations should be recorded. Truncating targets at ±0.1 means neither network learns a valid distance field away from the surface, ruling out sphere tracing. The budget is matched in iterations rather than in time, SIREN taking 63.6 seconds against the baseline's 56.2 for the same 5,000 iterations. The figures from the earliest superseded run were not retained, so Section 4.5 records that run's direction but cannot report it quantitatively.

# 6. Project Management and Reflection

## 6.1 Plan versus actual

The estimates held for the stages with a known shape, the literature review and the pipeline implementation both completing within their planned windows. They held poorly for the experimental stage, which was planned as a single block of runs and became three cycles of running, discovering a defect, correcting it and rerunning. The underlying error was treating experimentation as execution rather than as an iterative activity with its own failure rate. A second underestimate was extraction cost, which at 256 cubed takes between three and five times as long as training, so each rerun of a configuration cost close to six minutes rather than the minute the schedule assumed.

## 6.2 Deviations, risks and mitigations

Three risks from the Terms of Reference materialised. Memory exhaustion in Colab occurred during grid evaluation and was mitigated by evaluating the network in chunks and by developing at a grid resolution of 64 before committing to 256. Source mesh geometry was a second, since an inside test is only well defined on a closed surface and unreliable signs produce unusable training data, mitigated by an explicit watertightness check and repair before sampling. Dependency conflicts and loss of session state in Colab cost several hours of repeated setup, mitigated by caching the sample set to disk, which also turned out to be what made the two architectures comparable. The risk that did not materialise was training time, which the Terms of Reference identified as the likely bottleneck and which proved trivial next to extraction.

## 6.3 Reflection

The most useful thing learned came from the defects in Section 4.5. The direction of the headline result changed during correction, which means that had the defects gone unnoticed the dissertation would have reported a different conclusion with equal confidence. The third defect is the one worth reflecting on, because clamping the prediction as well as the target silently removed the gradient while the loss curve continued to look plausible, and it was found only because extraction failed on a field with no zero crossing. A metric that looks reasonable is not evidence that anything is working.

Given the project again, two changes follow. The first is one parameterised training entry point rather than two scripts, since duplication caused two of the three defects. The second is

instrumenting what a controlled comparison depends on rather than what is convenient to plot, which here meant gradient flow and sample distribution rather than loss alone.

# 7. Conclusion

## 7.1 Summary and answers to the research questions

Neural signed distance fields round the sharp features that make an asset usable, and this project tested whether the activation function is responsible by comparing a SIREN against a ReLU multilayer perceptron of identical capacity under a matched training budget, evaluated with metrics sensitive to localised rather than average error.

RQ1 receives a qualified yes. At the matched learning rate SIREN is better on every reported metric except the medians and maximum completeness, where the differences are small, but against the better of the two baseline rates tested the advantage narrows to the accuracy tail and to normal consistency, disappears at the median and reverses in completeness.

RQ2 finds that failures are contiguous rather than diffuse, forming between 35 and 39 connected regions in every configuration and concentrating on the same thin high curvature areas at roughly twice the curvature of the covered surface. The architecture changes the severity of these failures and not their location, which implicates the sampling and extraction stages rather than the nonlinearity.

RQ3 finds that symmetric Chamfer distance separates the two best configurations by 0.08 per cent while their accuracy tails differ by a factor of 3.3, because the aggregate sums two directed errors that here move in opposite directions and cancel. Aggregate and tail metrics disagree on which architecture is better, and the aggregate is the less informative of the two for this question. The hypothesis is therefore supported in its tail component, while its claim that the advantage is obscured in aggregate measures holds only against the baseline at 1e-3.

## 7.2 Contributions revisited

The pipeline and the controlled comparison deliver what Section 1.5 claimed, within the limits of one mesh and one seed. The evaluation protocol and the demonstrated divergence between aggregate and tail metrics are the more transferable contribution, since that divergence is a property of the metric rather than of this surface. The comparison itself supports a statement about these two architectures on this geometry at this budget and not a general claim about periodic activations.

## 7.3 Future work

Five experiments follow directly. Repeats across seeds would establish which differences here exceed initialisation variance. A CAD test set would test the crease and corner preservation

the introduction motivates and this mesh does not. Extraction at 512 cubed would separate the grid ceiling from network failure in the regions that fail. A screened Poisson baseline would anchor the results to production practice. A sweep over omega_0, and a hash grid encoding as a third architecture, would test whether detail is better obtained from the activation or from the input representation.

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# Appendices

Appendices A and B (Terms of Reference and ethics screening forms) are omitted from this public copy.

## Appendix C: Full hyperparameter tables

| Parameter | Value | Section |
|---|---|---|
| Source Mesh | Male Base Mesh, Free3D | 3.3 |
| Decimation Target | 50,000 faces | 3.3 |
| Final mesh | 24,461 vertices, 48,918 faces, watertight | 3.3 |
| Normalisation | Centroid at origin, bounding radius 0.9 | 3.3 |
| Total samples | 250,000 | 3.3 |
| Near surface samples | 200,000 | 3.3 |
| Uniform samples | 50,000 | 3.3 |
| Displacement scales | ±0.01 and ±0.05 per axis, half each | 3.3 |
| Negative sample fraction | 35.2% | 3.3 |
| Distance query chunk size | 1,000 points | 3.3 |
| **Architecture** | | |
| Input dimension | 3 | 3.4 |
| Hidden width | 256 | 3.4 |
| Hidden layers | 4 | 3.4 |
| Output dimension | 1 | 3.4 |
| Total parameters | 198,657 | 3.4 |
| Positional encoding | None, 0 frequency bands | 3.4 |
| SIREN omega_0 | 30 | 3.4 |
| SIREN first layer init | Uniform ±1/in_features | 3.4 |
| SIREN hidden init | Uniform ±√(6/in_features)/omega_0 | 3.4 |
| SIREN output init | Uniform ±√(6/hidden_dim)/omega_0 | 3.4 |
| Baseline init | Kaiming normal, variance 2/fan_in, biases zero | 3.4 |
| Baseline output init | Normal, standard deviation 0.1·√(2/hidden_dim) | 3.4 |
| **Training** | | |
| Loss | L1 | 3.5 |
| Target clamp | ±0.1 | 3.5 |
| Prediction clamp | None | 3.5 |
| Optimiser | Adam, default momentum parameters | 3.5 |
| Batch size | 256 | 3.5 |

| | | |
|---|---|---|
| Iterations | 5,000 | 3.5 |
| Sample draws | 1,280,000, or 5.12 effective epochs | 3.5 |
| Learning rate | 1e-4 both networks, 1e-3 baseline variant | 3.5 |
| Schedule, decay, clipping, early stopping | None | 3.5 |
| Seed | 0, NumPy and PyTorch | 3.5 |
| **Extraction** | | |
| Grid resolution | 256, development runs at 64 | 3.6 |
| Domain | [-1, 1] | 3.6 |
| Grid spacing | 0.00784 | 3.6 |
| Network evaluations | 16,777,216 | 3.6 |
| Marching cubes level | 0 | 3.6 |
| **Evaluation** | | |
| Surface samples per mesh | 200,000, fixed seed | 3.7 |
| Percentiles reported | 50, 90, 99, maximum | 3.7 |
| Coverage tolerances | 10 values from 0.002 to 0.030 | 3.7 |

## Appendix D: Superseded runs and corrections

| Defect | What the code did | How it was detected | What it invalidated |
|---|---|---|---|
| **Independent sampling** | Each script generated its own sample set with no shared seed, and the baseline's near surface perturbation displaced points inward rather than symmetrically | Each run of the code cell and training resulted in slightly different results and that randomness had to be eliminated for proper testing and documenting | All comparisons before the fix, since the networks trained on different and differently distributed data |
| **Held out split** | The baseline reserved 5% of samples, so it trained on 237,500 points against SIREN's 250,000 | Found by inspecting the sample count loaded by each configuration | The matched budget claim, since the sample counts differed |
| **Prediction clamped** | The L1 loss clamped the prediction as well as the target, so a network drifting | Extraction failed on a field containing no sign change | Any run where the network drifted, undetectably from the loss curve |

| | | | |
|---|---|---|---|
| | beyond ±0.1<br>received no gradient | | |

## Appendix E: Software documentation

The implementation consists of 1 Colab notebook: `MSc_SDF.ipynb`, which gets any .obj input and trains a SIREN and ReLU based MLPs with SDFs, for reconstructing the initial given mesh with custom resolutions.

All work was carried out in Google Colab on a CPU runtime. The versions were obtained from the environment used for the reported runs and listed below. Where a package was the Colab default it was not pinned, so a reader reproducing this work should install the listed versions explicitly rather than relying on the current defaults, since Colab's preinstalled versions change over time.

Python 3.13.15
numpy 2.1.3
scikit-image 0.25.2
scipy 1.16.3
torch 2.11.0+cpu
trimesh 5.1.0

How to run the program:
Insert a .obj file in the colab and then run each code cell by the sequence they are presented. Code cells with the comment “OPTIONAL”, are cells that just change the resolution of the model without retraining the whole model. Were made to save time in testing and early development

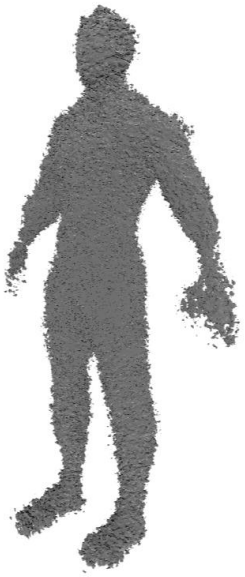
## Appendix F: Additional renders

These were structural tests on the SIREN MLP before implementing ReLU for comparison.

- 6 hidden layers:



- 512 MLP width:



- 512 and 6 hidden layers:

